GLANCE

The Ray-Ban Meta Smart Glasses integrate augmented reality, into eyewear, blending fashion and technology, The project seeks to redefine eyewear, offering a captivating experience. AI smart glasses use high-quality camera specifications to capture images, from social media feed posting to live streaming, health monitoring, speakers, and AR-based navigation, providing a multifunctional wearable for enhanced daily activities.

MIS 573

Project and Change Management

Meta Ray-Ban: AI Smart Glasses

.

By

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PART A : INTRODUCTION

A.1: Why is the project being developed

The Ray-Ban Meta Smart Glasses project represents a convergence of fashion and technology. The development of these smart glasses is motivated by the desire to revolutionize the eyewear market by seamlessly integrating cutting-edge augmented reality (AR) technologies into the iconic Ray-Ban spectacles.

With a focus on creating a unique and engaging user experience, the project aims to redefine the traditional notion of eyewear by offering consumers a blend of elegance and utility. By combining Meta's advanced technology capabilities with Ray-Ban's timeless style, the goal is to provide users with a distinctive wearable device that not only enhances their daily activities but also reflects their personal style.

The project's inception was fueled by the need to adapt to the changing landscape brought about by the COVID-19 pandemic, which rendered Meta's testing labs inaccessible. Engineer Igor Markovsky's ingenuity in setting up a rapid design and validation station in his home enabled the team to continue their work despite the challenges posed by the lockdown.

The collaboration between Meta and Essilor Luxottica presented a significant challenge in re-engineering components to ensure that advanced features such as dual cameras, micro-speakers, audio arrays, a powerful Snapdragon® processor, a capacitive touchpad, and a battery could be seamlessly integrated into the lightweight and stylish Ray-Ban frames.

Ultimately, the Ray-Ban Meta Smart Glasses project seeks to advance wearable technology by offering consumers a product that not only meets their technological needs but also complements their personal style. With its emphasis on innovation, user-centric design, and seamless integration, the project aims to pave the way for a revolutionary journey in the field of augmented reality and wearable technology**. (Loh & Misselhorn, 2019)**

A.1.2 : Unique Value Proposition

During the summer of 2020, Meta engineers working on Ray-Ban Stories, the company's first smart glasses that allow you to create video and images, share your life's memories, listen to music, and answer calls, encountered an issue. The glasses needed a charging case, however, none of the designs worked adequately. Combining state-of-the-art augmented reality (AR) technology with the enduring quality and design of the Ray-Ban brand creates a distinctive value proposition for the Meta Ray-Ban. **(Slotwiner, 2023)** The following salient features accentuate its distinct worth:

Augmented Reality Integration: Meta Ray-Ban skillfully incorporates augmented reality technology with fashionable eyewear, offering consumers a dynamic and engaging experience. By enabling hands-free information access, environment navigation, and digital content interaction, users can increase their productivity and enjoyment.

Iconic Design: Ray-Ban eyewear has become a symbol of sophistication and fashion because to its timeless design components, which Meta Ray-Ban has maintained from its founding on the brand's well-known style and quality. People who value both style and utility will find this fashion and technology combine appealing.

A.1.3: Market Competitors: compare it to similar products/services

Competitors of the Ray-Ban Meta smart glasses include various consumer tech manufacturers, such as the headphone and speaker company Bose, the gaming equipment company Razer and the computer company Lenovo and many more.

- **Google Glass:** Among the earliest widely known smart eyewear products was Google Glass. Despite significant obstacles to consumer adoption in its first iteration, Google is still researching and developing AR glasses for a range of uses, including business and industrial use cases.
- **Microsoft HoloLens:** HoloLens is a mixed reality headset that is aimed mostly at business users. Competing in the larger AR glasses market, it has strong capabilities for applications including design visualization, remote support, and training.
- **Snapchat's parent company, Snap Inc.,** has unveiled many versions of its smart eyewear, called Spectacles. Although Snap Spectacles were once designed to capture and share images and videos, they are now including augmented reality (AR) features and experiences, making them rivals in the consumer smart eyewear market.
- **Apple AR** glasses are said to be in development, albeit they haven't been officially announced as of the time of my previous post. Apple's debut into the smart eyewear industry is highly anticipated and might offer considerable competition to existing vendors due to its strong brand presence, ecosystem integration, and focus on user experience**.**
- **Samsung Smart Glasses**: Samsung has experimented with concepts and prototypes of smart eyewear in the past, but it hasn't introduced a commercially accessible model in this area. Samsung is still a possible rival in the smart glasses sector, though, considering its technological prowess and market presence.
- **Vuzix:** This company specializes in smart glasses and augmented reality technology. Positioning

itself as a competitor to other players in the smart eyewear area, it provides a range of solutions targeted at consumer, enterprise, and industrial industries.

A1.4: Strategic Narratives

We see smart glasses as a stepping stone to full augmented reality glasses, with an emphasis on creating something you'll want to wear regularly for lengthy periods. We also believe people must become accustomed to using smart glasses in everyday life (for example, taking a video while skiing, taking a photo of your dog while you play together, or listening to music while running) for AR glasses to become an essential technology one day.

Immersive Learning Environment: Users can interact with digital content superimposed on the real world through the Meta Ray-Ban's immersive augmented reality experience. To create extremely engaging learning experiences where users can visualize complicated concepts, interact with virtual objects, and take part in simulations—all directly through their glasses—is what the project team focusing on education and training is tasked with.

Tailored Learning Paths: By utilizing Meta Ray-Ban's adaptable functionalities and customized augmented reality encounters, the research group can customize instructional materials to suit individual learning preferences and patterns. The team may enhance learning outcomes by adjusting the learning environment in real time through the analysis of user behavior and interactions, along with targeted feedback and recommendations.

Possibilities for Collaborative Learning: By enabling several users to interact with the same augmented reality environment at once, Meta Ray-Ban's connection features facilitate collaborative learning experiences. Collaboration and information sharing among students are encouraged by the chances this creates for group projects, peer-to-peer learning, and group-based activities. **(Iqbal & Campbell, 2022)**

Integration with Current Platforms: The Meta Ray-Ban simplifies the process of creating and implementing augmented reality apps by integrating smoothly with current platforms and development tools as part of Meta's ecosystem. By doing away with the technological difficulties of starting from scratch, the project team can concentrate on creating engaging content and educational opportunities. **(Iqbal & Campbell, 2022)**

To summarize, the Meta Ray-Ban presents a perfect platform for the project team to realize their objectives of utilizing immersive augmented reality experiences to transform training and education. Learners are empowered to explore, participate, and learn in ways that were previously unthinkable with the Meta Ray-Ban's advanced features, user-friendly interface, and seamless integration capabilities.

A1.5: Project Goals

Our project aims to create smart glasses that enhance your daily experience without interfering with your privacy or connecting you to the outside world. We want them to be long-lasting, fashionable, comfortable, lightweight, and packed with high-tech features. Your routine should be effortlessly integrated with the glasses.

PART B : PROJECT CHARTER

The Project Charter helps in project planning and used as a guide throughout the project's lifespan. It permits the project manager to utilize organizational resources to carry out the project's goals. The project charter for "Ray-Ban Meta Smart Glasses" is shown below:

B.1 Project Objectives:

The objectives for Ray-Ban Meta Smart Glasses are as follows:

- **Innovation and Technology advancement**: The development of smart glasses requires the integration of Meta's advanced AI, Voice recognition, and Software and machine learning algorithm for photo capturing Augmented Reality.
- **UX experience**: The meta glasses allow users to capture photos and videos, listen to music, and connect to social media platforms by using only smart glasses.
- **Advanced features**: The second-gen Ray-Ban smart glasses feature Al assistance like Siri, and Meta, to connect with users aiming to elevate the user's overall experience, to help in content creation and other day-to-day activities.
- **Frame-Design:** To create frames that are modern and stylish in looks and able to create customized frames and glasses to their fullest.

In-order to overcome the drawbacks of the Ray-Ban stories (1^{ST} Generation), the additional features have been embedded to make the smart glasses more user friendly.

- **Camera Specifications:** The new Meta| Ray-Ban smart glasses provide several notable upgrades to their camera from 5MP to 12 MP, and 1080p video to meet customer's demands for higher quality visuals.
- **Ray-Ban Meta Smart Glasses:** packing an ultra-wide camera, support 12MP photos and 60-second, 1080p videos. **(Iqbal & Campbell, 2022)**
- **Social Interactions:** The new Qualcomm Snapdragon AR1 GEN 1 platform for faster response,

with Wi-Fi 6 and Bluetooth 5.3 standards we can live stream through smart glasses from a first-person perspective with a wide angle. And we can see the comments in your preview or tap and hold on the side of your glasses to hear them out loud.

- **Smart Controls:** Unlock a range of hands-free possibilities with voice commands – call and message, capture content, and manage media settings with your phone in your pocket. Or activate the features in a single touch with the hyper-responsive touchpad and capture button.
- **Durability:** The smart glasses are designed to withstand extreme weather and water resistance to prevent accidental fall damage.
- **Meta VIEW:** with the launch of the meta companion app every user can access their amazing journey of photos and videos saved in storage, even it is beta stage the application is compatible with most smartphones.

B.2. Scope for Ray-Ban Meta Smart Glasses:

With smart glasses, easily merge the digital and physical worlds to transform how people interact with the environment. The scope of work for the development of smart glasses like Ray-Ban Meta Glasses would involve various aspects, including:

- **Hardware Development**: Design and manufacturing of the physical glasses, including the integration of display technology, sensors, cameras, and other hardware components.

- **Software Development:** Development of the operating system, user interface, and applications that run on the smart glasses. This may also involve creating software development kits (SDKs) for third-party developers to create applications.

- **Connectivity**: Integration of wireless communication technologies like Bluetooth or Wi-Fi to enable connectivity with other devices such as smartphones, tablets, or the internet.

- **User Experience (UX) and User Interface (UI) Design:** Creating an intuitive and user-friendly interface for users to interact with the smart glasses. This includes designing the visual elements and ensuring a seamless user experience.

- **Battery Management:** Addressing the challenge of optimizing battery life, as smart glasses typically require power for various functions and should offer reasonable usage time on a single charge.

- **Durability and Comfort:** Ensuring that the glasses are durable, lightweight, and comfortable for extended use.

- **Compliance and Regulations:** Adhering to regulatory requirements and standards related to eyewear and technology products in different regions.
- **Marketing and Launch:** Planning and executing marketing strategies for the product launch, including promotional activities and partnerships.
- **Health and Fitness Monitoring:** Monitor indicators related to health and fitness.
- **Customer Support and Maintenance:** Establishing customer support channels and processes to address user queries and issues. Additionally, providing software updates and maintenance support.

B.2.1: Roadmap:

Roadmaps are the output of a strategic planning process. You can link goals to detailed work and show the time frame for achievement, given your resources and capacity

Phases	Deliverable
Phase 1	Roadmap aims to develop the essential features and launch for early adopters.
Phase 2	Based on users feedback, and stakeholder inputs, new additional functionality to be incorporated to lure more customer base.
Phase 3	Combine with more Meta goods and services to form a whole ecosystem.

B.2.2 Success Metrics

Success metric is a quantifiable measurement that Managers track to see if their strategies are working effectively:

- Engagement and adoption rate of users.
- Development and integration of apps.
- Favorable effects on the businesses and applications in question.
- Taking care of issues and developing responsibly.
- This scope lays the groundwork for future development and improvement by concentrating on the key components of Ray-Ban Meta smart glasses.

B.3 Milestone

Project milestones are checkpoints in your plan that mark important events, dates, decisions, and deliverables so it's easy for your team and stakeholders to map forward progress on the project.

Milestone	Deliverable	Date
Project Implementation	Project Implementation **Justification**: Develop the draft of the project based on the requirements of Stakeholders and sponsors to create project objectives and scope of the project. **Explanation**: define the project objectives and scope to understand the software and hardware requirements to set and identify market trends in smart eyewear and their preferences.	15 Jan 2024
Research and development	Once the project is sanctioned by the sponsors and team manager, the team charter is created and signed. the development team will start working on research and outline the project aspects. The project is sanctioned and the research data and the identification of required resources for the project will help the team members create a common goal **Explanation**: After sorting ideas that can be implemented. Align design specifications to match quality, and appearance with the brand's fashion wear and user preferences.	11 May 2024
Initial prototype	**Justification**: Once the research and development phase is finished, the team will start with the prototype of the smart glasses and compare it with generation One [Ray-Ban Stories, and the development of a companion	1 August 2024

	Android, IOS application is required to access the contents created using smart glasses. **User's feedback and prototype improvement**: feedback received from the user who participated in the testing phase is taken into consideration, and the development team needs to recreate the prototype that passes all the.	
Overcoming drawbacks:	Justification: In this stage, the team will work to address any constraints or difficulties encountered during the initial launch of the smart glasses.	14 March 2025
Final launch:	**Justification**: The product is ready to go live and designed and manufactured, ensuring that it meets all the passes all the requirements. Conduct rigorous testing to ensure that the product is durable and meets all relevant safety and quality standards. **Explanation:** Maintenance and consistent software updates to stay competitive till the product life.	1st July 2025

B.4 Budget :

The below are the key factors responsible for the project budget estimation:

Sr no	Factors	Description
1	**Research and Development:**	**Justification**: To research more about the metaverse initiatives, technological advancements, and prototype development to analyze the functions and performance of the Meta Wayfarer glasses.

		Explanation: To be successful in the ever-changing Metaverse world, Meta Wayfarer glasses require research and development (R&D). To maximize functionality, performance, and user experience, R&D investigates cutting-edge AR/VR technology, studies Metaverse trends, and improves prototypes. This eventually drives the success of Meta Wayfarer glasses by ensuring competitive positioning, minimizing product risks, and maximizing user adoption within the dynamic Metaverse.
2	**Production and Manufacturing**:	**Justification**: Costs associated with sourcing materials, manufacturing processes, quality control, and assembly of the Meta Wayfarer glasses. **Explanation**: The budget allocated to production and manufacturing guarantees the financial sustainability of Meta Wayfarer glasses through precise pricing, process optimization, risk mitigation, and scalability facilitation. It takes into consideration the procurement of raw materials, production procedures, quality assurance, and assembly; in the end, these factors affect profitability, operational effectiveness, and our capacity to satisfy future market demands.
3	**Marketing and Promotion:**	**Justification**: For marketing campaigns, advertising, brand promotion, and retail support to generate awareness and drive demand for the Meta Wayfarer glasses. **Explanation**: Meta Wayfarer glasses are successful because of their marketing and promotion, which raises consumer awareness of the brand, creates buzz, and eventually increases sales. This budget guarantees that customers get the value proposition and are willing to embrace the Metaverse through these cutting-edge

		glasses through focused campaigns, strategic advertising, captivating brand storytelling, and effective retail partnerships.
4	**Distribution and Logistics**:	Distribution channels, transportation, and inventory management to ensure timely delivery of the Meta Wayfarer glasses to customers. **Explanation**: To ensure that Meta Wayfarer glasses are delivered to consumers without incident, a flawless distribution and logistics network are essential. This budget addresses the establishment of dependable and affordable transportation choices, the implementation of strong inventory management systems, and the creation of effective distribution channels (online and retail partners). Simplifying these processes would ensure that products are available on time, reduce delivery delays, and eventually improve customer happiness and brand recognition.
5	**Sales and Support**:	Budget for sales team resources, marketing, visualization development team, website development team, customer support infrastructure, training, and after-sales service to facilitate customer acquisition and retention. **Explanation**: Long-term success of Meta Wayfarer glasses depends on investments in sales and support. Through focused marketing efforts, a talented staff developing visualizations, and a committed sales force, this funding helps to accelerate customer acquisition. A strong website also makes it easier for customers to explore products, and continuing education, excellent after-sales care, and a vast customer support network guarantee that customers stay with the company and develop brand loyalty. An enthusiastic user base of

		Meta Wayfarers and steady sales growth are made possible by this all-encompassing strategy.
6	**Technology Infrastructure**:	Investment in IT systems, software platforms, and infrastructure upgrades to support the development, deployment, and maintenance of the Meta Wayfarer glasses and associated services. **Explanation**: Meta Wayfarer's complete experience is supported by a robust technology infrastructure. With the help of this budget, the glasses and their supporting services can be developed, implemented, and maintained. It is possible to guarantee reliable data protection, effective software development and maintenance, and seamless and secure data storage by investing in IT systems, software platforms, and infrastructure updates. As a result, Meta Wayfarer glasses will succeed and find a home in the developing Metaverse thanks to a smooth user experience, dependable functioning, and future

B.5 Success criteria:

Below are the success criteria for Ray-Ban Meta Glasses:

Definition	**Description**
People Comfort/Fit:	Justification: Meta Glasses stand out for their comfort and user-friendly design, whereas plays a pivotal role in enhancing user satisfaction and fostering widespread adoption of cutting-edge technology, these factors greatly impact user happiness and overall adoption of the new advancement technology. Explanation: The success of the project totally depends on user willingness to use the glasses happily, the essential key is to make the

	glasses fit seamlessly into users lives, ensuring they enjoy using them. A satisfied user base indicates a successful project.
User friendliness / Safety:	Justification: it is important to make sure that the glasses are easy to use and safe. It enhances employee productivity and safety. Productivity is increased If using the hands-free function of the glasses is simple. Explanation: To make the project successful, the glasses need to improve people's productivity and safety. For productivity, they must be simple to use and have hands-free capabilities.
Product Growth:	Justification: The revenues and growth of the smart glasses in interlinked with availability in multiple regions rather than specific locations. This can be achieved by expanding the distribution channels once the product receives trustworthy feedback from the customers. Moreover, introducing new privacy policies, and GPS enabling a location tracker in case of stolen or loss, will attract customers and increase company growth. Explanation: These strategies of marketing will create awareness and increase in popularity among the new generation of customers.
Technical Support:	Justification: Timely responses to any customer inquiries and resolving technical issues are crucial for building positive responses and loyalty to/from users. Explanation: Investing in knowledge-driven support staff and establishing efficient support channels is the key to providing prompt assistance to resolve user queries. Furthermore, Automation and enhanced self-service customer support will further enhance the process.

B.6 Approach:

Sr no	Type	Justification	Explanation
1	User-Centric Redesign	Justification: Objective: The aim of the justification is prioritizing user happiness for increased satisfaction and product usage. Strategy: To improve user happiness, direct user feedback on appearance, hand-free use, and sound, targeting key areas for user satisfaction. Implementation: To make sure a positive and pleasurable user experience Implement changes based on user input to improve looks, ease of use, and sound quality, ensuring a positive and enjoyable user experience.	Explanation: Objective: Improve the Ray-Ban Meta smart glasses by focusing on what users needs and ensuring they are satisfied with the new look. Strategy : Focus on the appearance of the glasses, on how the glasses look, being hands-free, and having good audio. This redesign will be aligned with what user's value most. Implementation: make alter according to the design based on what users say, concentrating on improving how the glasses handle media and making sure that they are comfortable to wear it. This makes the redesigned glasses more enjoyable and practical for users.
2	Audio Enhancement	Objective: Better audio contributes to users happier with the device, primarily people generally like clear and immersive sound. Strategy: The demand for higher volume audio is immediately satisfied	Objective: Improve the audio experience better for users to increase overall satisfaction with the device. Strategy: Develop special speakers with better bass,

		by designing speakers with better directional audio deeper bass and more volume Implementation: Adding a new microphone setup supports better audio for recording videos, making it useful for users who create content. This covers both listening and recording needs for a more versatile audio experience.	higher volume, and improved directional audio to directly address what users actually need in audio quality. Implementation: Adding a new microphone setup to support immersive audio recording during video capture, ensuring high-quality audio in different environments.
3	Advance imaging Capabilities:	Objective: Enhancing the image is the key for user satisfaction through better images and videos. Strategy: Introducing an ultra-wide 12 MP camera and 1080p video to align with the user demands for higher quality visuals. Implementation: The integration of Voice-activated sharing enhances usability, making advanced imaging features user-friendly for seamless experiences.	Objective: Improve photos and videos through camera upgrades to boost consumer satisfaction. Strategy: Integrate an ultra-wide 12 MP camera to improve photo and 1080p video quality, meeting user preferences. Implementation: To provide even more ease, voice command sharing for added convenience, making it easier for users to share photos seamlessly.
4	Technologies Advancements:	Objective: Using the latest technology ensures the device performance really well, meeting what users want for advanced functionality. Strategy: Qualcomm Snapdragon AR1 Gen1 Platform power the glasses, satisfying the user demands for high-	Objective: we aim to have a high performance device with advanced features users want . Strategy: By using Qualcomm Snapdragon AR1 Gen1 platform

		quality photo and video processing in augmented reality applications. Implementation: Designing a sleek charging case that can hold eight additional charges ensures 36 hours of use, improving user convenience and meeting the need for sustained performance.	Implementation: We meet the user demands for top quality photos and videos processing in augmented reality, in addition to this a sleek charging case with eight charges ensures 36 hours of use.
5	AI Implementation	Objective: Improving user interaction through advanced conversational support is essential for a smooth and simple experience. Strategy: Including Meta AI on Ray-Ban Meta smart glasses provides a hands-free on-the-go- experience while meeting user expectations for cutting-edge technology. Implementation: Optimizing Meta AI for voice commands like "Hey Meta" provides an interaction and efficient way for users to access information, enhance creativity, and control features, connecting with the demand for advanced and hands-free interactions.	Objective: Conduct user survey users and test the integration of Meta AI integration on a subset, collecting feedback on satisfaction and perceived improvements. Strategy: Launch Meta AI smart glasses, monitor user engagement, and gather feedback on the overall experience and convenience. Implementation: Optimize Meta AI for voice commands like "Hey Meta." Test the usability of the product to gauge its efficiency and get user input for ongoing development.

B.7 Roles and Responsibilities

Roles And Responsibilities			
Name	Role	Position	Contact Information
1.Shrestha, Sudip Kumar	Project Manager, Marketing Manager,	Manager	sshre@uis.edu
2.Soni, Rahul	Team Member	Tech Led, User Experience Research, AI Specialist.	rsoni30@uis.edu
3.Bachina,Jagadish	Team Member	Design lead, Quality Assurance.	Jbach8@uis.edu
4.Pushkekar, Rohith Kumar	Team Member	Engineering Team, Legal and Compliance	rpush6@uis.edu
5.Bandaru, Pavan Kalyan	Team Member	Sales Team, Customer Support, Supply and Chain Management.	pband6@uis.edu

B.7.1 Roles Description:

Roles	Responsibilities
Product Manager	Ensuring that everything is going according to the plan and what user's needs.
Design Lead	A design lead is responsible for providing creative direction, managing design teams, overseeing projects, maintaining, and setting design standards, supervisor projects, establishing and upholding design standards.
Engineering Team:	The team responsible for ensuring that the insides of the glasses function properly, including cool features and general maintenance.

Marketing Manager	Accountabilities: Promotes the eyewear, organizes sales strategies, and organizes launch events.
Sales Team	Organizes activities for the debut of the glasses, informs everyone about them, and develops the sales strategy.
Customer Support:	Helps customers with queries or issues they may have both before and after purchasing the glasses.
Quality Assurance:	Checks the quality of the glasses and makes any necessary adjustments.
Legal and Compliance:	Ensures that the glasses abide by all regulations and takes care of any legal concerns.
Supply Chain Management:	Makes sure there are enough glasses for sale and makes timely deliveries of them.
AI Specialist:	Enhances the overall experience and adds intelligence to the glasses by developing capabilities that comprehend human input.

B.8 Project Assumptions

Project assumptions are part of every project management cycle. Project assumptions are the expectations and predictions that a team assumes when they start a project.

Sr no	Assumption	Description
1	Schedule	Materials will arrive as planned within the project schedule Vendor contracts will be fully executed within two months of vendor selection.
2	Methodology	Project will follow waterfall methodology throughout execution Project will follow team governance guidelines and requirements
3	Budget	Project costs will stay the same as initially budgeted costs Training will be conducted internally with no additional training costs incurred
4	Regulatory Compliance:	To maintain user safety and legal compliance, the project expects that Meta Smart Glasses will abide by government-stated regulations and standards on eyeglasses and technology products.
5	Collaboration Continuity	It is anticipated that our project can be collaborated with other industries for investment, research and development making it possible to seamlessly incorporate cutting-edge technology into the Meta Smart Glasses.
6	Incorporation of User Feedback	It is assumed that input from users during the prototype stage will be considered and integrated into the development process to make sure the end product meets user needs and expectations.
7	Milestone Achievement	It is assumed that the project's milestones—such as the first prototype, overcoming obstacles, and the final launch—will be completed within the allotted time frames, enabling the project to move forward smoothly.
8	Technological Landscape Stability	Because the project is predicated on a stable technological environment, state-of-the-art parts such as dual cameras, micro-speakers, battery performance and a potent processor will be integrated into the fashionable and lightweight frame.

Part C: Work Breakdown Structure and WBS Dictionary

C.1: Table View

The below table represent the WBS structure for the Ray-Ban Meta Smart Glasses along with the WBS dictionary.

Work Breakdown Structure for Ray-Ban Meta Smart Glasses (Table View)			
Level 1	**Level 2 (Work Package)**	**Level 3**	**WBS Dictionary**
1. Ideation			The work to initiate the project.
	1.1 Project Initiation	1.1.1 Market Research and Analysis	This helps the business understand the competition and threats and discover advantages and opportunities in the market.
		1.1.2 Develop Project Charter	It helps in project planning and may be used as a guide throughout the project's lifespan. It permits the project manager to utilize organizational resources to carry out the project's goals.
		1.1.3 Project Charter Signed/Approved	Project Sponsor reviews the project charter and give confirmation
2. Project Management			Planning and estimation of project.
	2.1 Project Planning	2.1.1 Create a Scope Statement	The project scope defines the boundaries of the project, including what will be included and what will not. This is important to ensure that the project stays focused and on track

Work Breakdown Structure for Ray-Ban Meta Smart Glasses (Table View)			
Level 1	**Level 2 (Work Package)**	**Level 3**	**WBS Dictionary**
		2.1.2 Develop a Communication Plan	It lays out precisely how you're going to accomplish your goals and explains how you're going to quickly and effectively communicate it to the other team members.
		2.1.3 Develop Success Metrics	It is imperative to develop metrics to evaluate the progress of the project.
		2.1.4 Develop Project Schedule	It outlines the project's timeframe, resources, and reality of completion.
		2.1.5 Identify Project Team	It assists project managers in assigning tasks efficiently and scheduling the ideal team for the project.
		2.1.6 Develop Budget Plan	It enables the project manager to determine his financial limitations for each individual project phase.
		2.1.7 Develop Staffing Plan	It helps define how the project will utilize human resources throughout its entire lifespan.
	2.2 Meetings	2.2.1 Organize Kickoff meeting	The purpose of this meeting is to start out well and have everyone on the same page.
		2.2.2 Status Meeting	Status meetings boost team morale, offer timely updates on the project, reveal potential risks, offer a platform for prompt problem resolution, and promote the sharing of pertinent information.

Work Breakdown Structure for Ray-Ban Meta Smart Glasses (Table View)			
Level 1	**Level 2 (Work Package)**	**Level 3**	**WBS Dictionary**
Requirement and elicitation of project.			Requirement and elicitation of project.
	3.1 Architecture Requirements (Hardware)	3.1.1 Draft Requirements	The requirements outline precisely what the Ray-Ban Meta glasses features i.e. software and hardware specification and the acceptance criteria's.
		3.1.2 Requirement document Approval	Approval of the requirement document is a review process involving relevant stakeholders such as product managers, engineers, and quality assurance teams. Stakeholders review it and share feedbacks and document revisions are made before final approval, to make sure the product goal are met.
	3.2 Mobile App Architecture Requirements (Software)	3.1.1 Draft Requirements	This requirement outline the features of mobile application and architecture details
		3.1.2 Requirement document Approval	Stakeholders review it and share feedbacks and document revisions are made before final approval, to make sure the product goal are met.

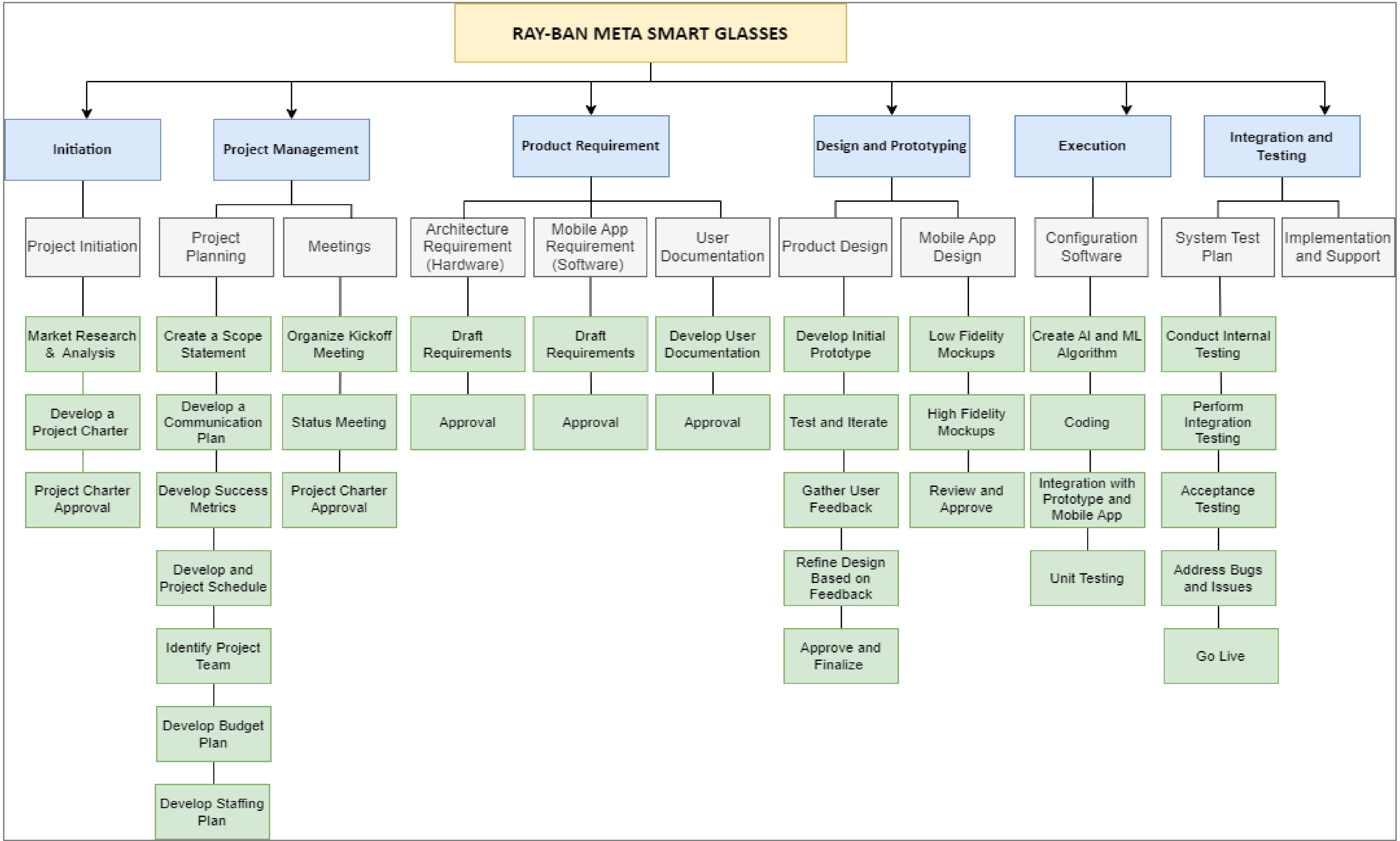
Work Breakdown Structure for Ray-Ban Meta Smart Glasses (Table View)			
Level 1	**Level 2 (Work Package)**	**Level 3**	**WBS Dictionary**
	3.3 User Documentation	3.2.1 Develop User Documentation	We provide clear and concise documentation on how our product is to be used. Here we include the introduction of our product its features and benefits how to install the software and hardware manual, a description on how to perform basic operations, it includes any advanced features and glossary or
		3.2.2 Approve User Documentation	Approval of documentation is an iterative process with multiple iterations of reviews, feedback, and revisions as necessary to ensure that the final product is of high quality and meets the needs of the target users.
4. Designing and Prototyping			Designing and prototype of project
	4.1 Product Designs	4.1.1 Develop Initial Prototype	It's an initial creation of a product that shows the basics of what a product will look like, what the product will do, and how the product operates
		4.1.2 Test and Iterate on Prototype	Testing of prototypes design and make the changes
		4.1.3 Gather User Feedback	Gathering feedback from stakeholders
		4.1.4 Refine Design Based on Feedback	Implement / make adjustments, and improve the overall user experience
		4.1.5 Approve and finalize	The select design is used by the software engineers as the wireframe for the development

Work Breakdown Structure for Ray-Ban Meta Smart Glasses (Table View)			
Level 1	**Level 2 (Work Package)**	**Level 3**	**WBS Dictionary**
	4.2 Mobile App Design	4.2.1 Design Low fidelity Design	This is the most basic visual representation of the user interface and is typically where the design process begins. This is drawn using tools like balsamiq or pencil.
		4.2.2 Design High fidelity Design	High-fidelity design have real featured pictures and pertinent text.
		4.2.3 Review and Approve	The select design is used by the software engineers as the wireframe for the development of mobile application.
5. Execution			Development phase of project
	5.1 Software Development	5.1.1 Configure Software	Defines the system's logical flow, inputs and outputs, data organization, relevant business, and processing rules, and how it should appear to users.
		5.1.2 Creating AI and ML Algorithm	To understand the pattern of human behaviour, identify objects and translate languages
		5.1.3 Coding	Involves writing programmable instructions as to how the app should work.
		5.1.4 Integration with prototype and mobile app	Will make it possible for the mobile app to have a feature that will give users, Meta glasses details in real time. Connectivity Features (Bluetooth, Wi-Fi, etc.)

Work Breakdown Structure for Ray-Ban Meta Smart Glasses (Table View)			
Level 1	**Level 2 (Work Package)**	**Level 3**	**WBS Dictionary**
		5.1.5 Unit Testing	It makes it easier to find and fix errors or flaws at the code level. This step tests every iteration to make sure it is working to avoid incurring massive technical debt.
6. Integration and Testing			Testing of project for bugs
	6.1 System Test Plan	6.1 .1 Conduct Internal testing	This is whereby the designed prototype and mobile app are individually shared to a small number of people to use it. The goal is to detect or expose technical flaws or errors in one or more components.
		6.1.2 Perform Integration Testing	This is to test how the Glasses AI and ML and mobile app interact and get the job done.
		6.1.3 Acceptance Testing	Enables an organization to engage end users in the testing process and gather their feedback to relay to developers.
		6.1.4 Address Bugs and Issues	Address all the bugs and issue that occur at the time of testing phases
		6.1.5 Go live	Moving from development environment to production environment and it goes live to all users
	6.2 Implementation and Support		Support and maintenance thorough out the project period

C.2 : Tree View

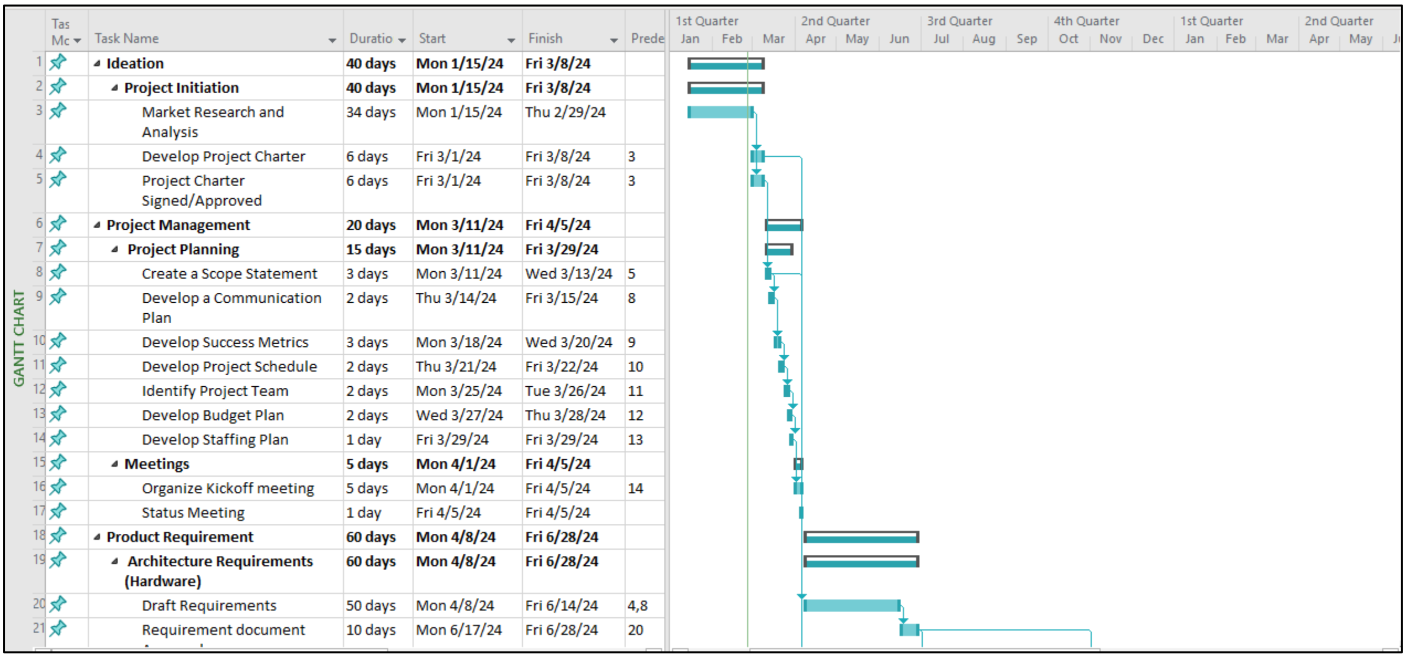
The below diagram shows the tree view of the Work Breakdown Structure. Tree view helps in making it easier to visualize the easily and Fastly.



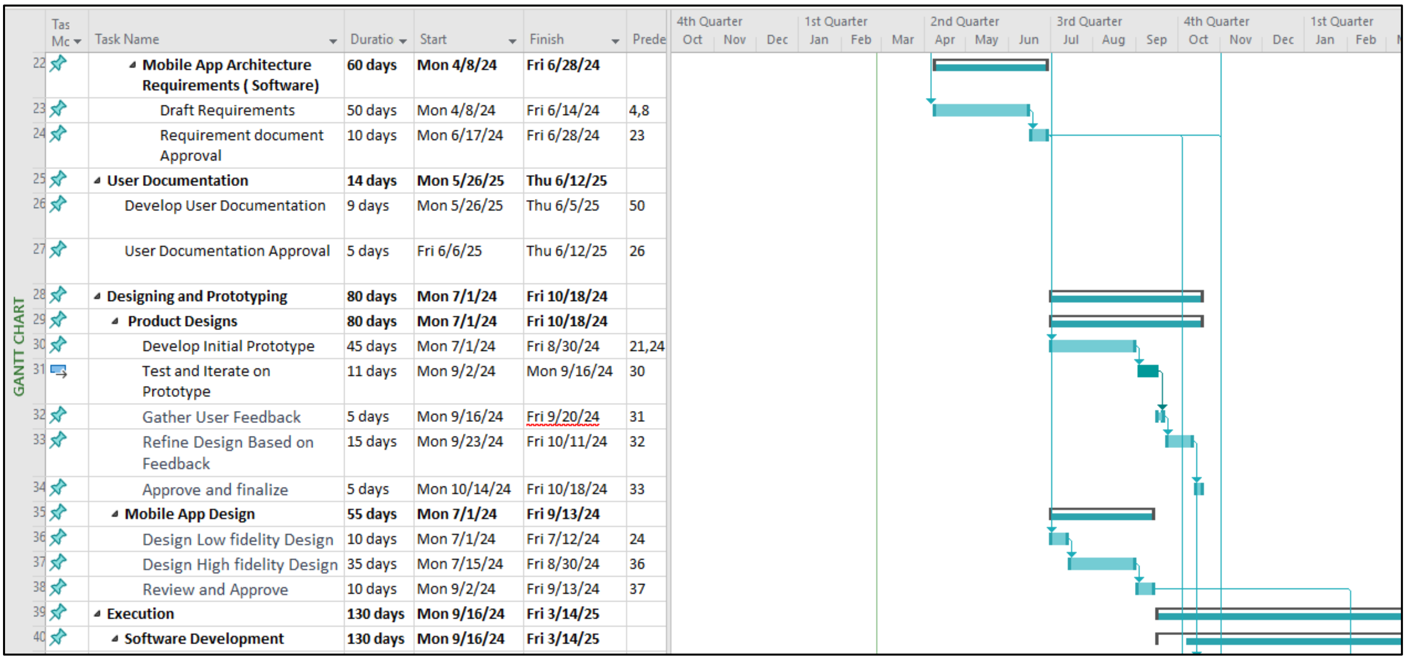
PART D : GANTT CHART

A Gantt chart's critical path is the order of jobs that establishes the project's earliest possible completion date. This path is essential because any delays in completing tasks will push back the deadline for finishing the entire project. The length of each task and the dependencies between tasks are examined to identify the critical path. The critical path, which is the longest path in the project network diagram, denotes the amount of time that must pass for the project to be finished

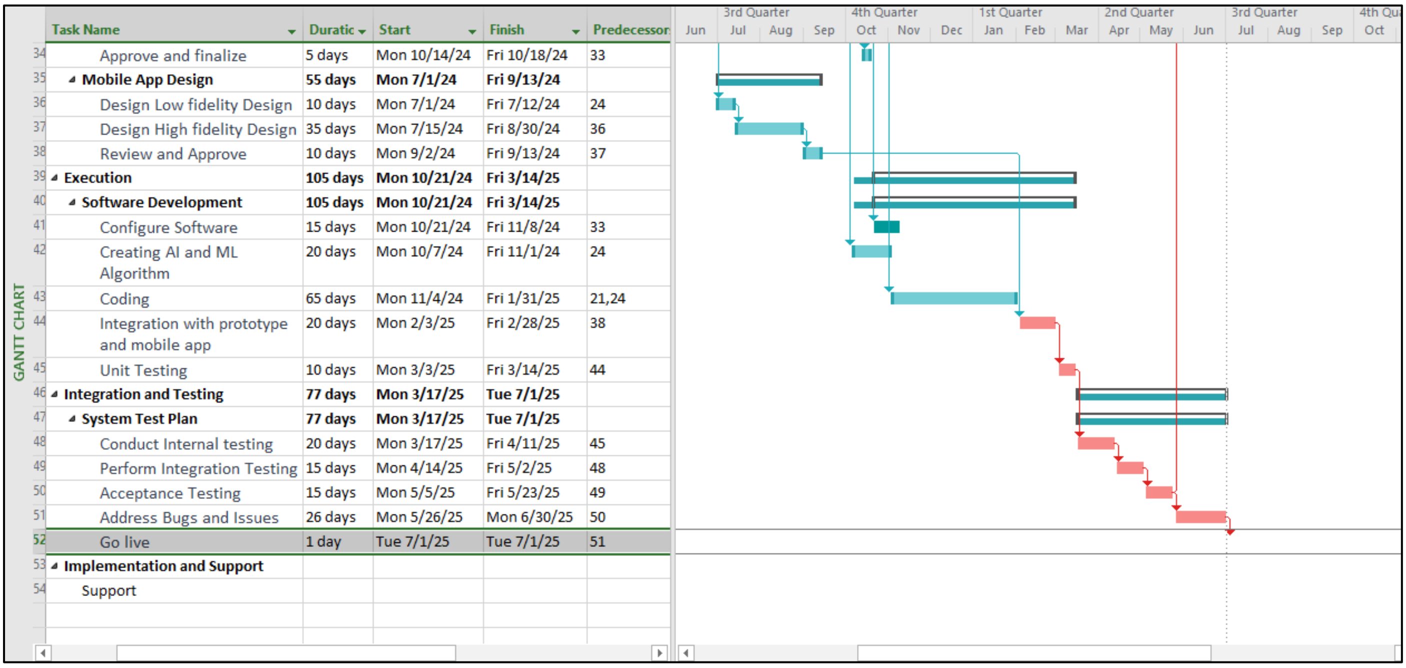
For identifying the critical path we need to identify the predecessors for each activity.



	Task Name	Duration	Start	Finish	Predecessors
1	**Ideation**	**40 days**	**Mon 1/15/24**	**Fri 3/8/24**	
2	**Project Initiation**	**40 days**	**Mon 1/15/24**	**Fri 3/8/24**	
3	Market Research and Analysis	34 days	Mon 1/15/24	Thu 2/29/24	
4	Develop Project Charter	6 days	Fri 3/1/24	Fri 3/8/24	3
5	Project Charter Signed/Approved	6 days	Fri 3/1/24	Fri 3/8/24	3
6	**Project Management**	**20 days**	**Mon 3/11/24**	**Fri 4/5/24**	
7	**Project Planning**	**15 days**	**Mon 3/11/24**	**Fri 3/29/24**	
8	Create a Scope Statement	3 days	Mon 3/11/24	Wed 3/13/24	5
9	Develop a Communication Plan	2 days	Thu 3/14/24	Fri 3/15/24	8
10	Develop Success Metrics	3 days	Mon 3/18/24	Wed 3/20/24	9
11	Develop Project Schedule	2 days	Thu 3/21/24	Fri 3/22/24	10
12	Identify Project Team	2 days	Mon 3/25/24	Tue 3/26/24	11
13	Develop Budget Plan	2 days	Wed 3/27/24	Thu 3/28/24	12
14	Develop Staffing Plan	1 day	Fri 3/29/24	Fri 3/29/24	13
15	**Meetings**	**5 days**	**Mon 4/1/24**	**Fri 4/5/24**	
16	Organize Kickoff meeting	5 days	Mon 4/1/24	Fri 4/5/24	14
17	Status Meeting	1 day	Fri 4/5/24	Fri 4/5/24	
18	**Product Requirement**	**60 days**	**Mon 4/8/24**	**Fri 6/28/24**	
19	**Architecture Requirements (Hardware)**	**60 days**	**Mon 4/8/24**	**Fri 6/28/24**	
20	Draft Requirements	50 days	Mon 4/8/24	Fri 6/14/24	4,8
21	Requirement document	10 days	Mon 6/17/24	Fri 6/28/24	20



	Task Name	Duration	Start	Finish	Predecessors
22	**Mobile App Architecture Requirements (Software)**	**60 days**	**Mon 4/8/24**	**Fri 6/28/24**	
23	Draft Requirements	50 days	Mon 4/8/24	Fri 6/14/24	4,8
24	Requirement document Approval	10 days	Mon 6/17/24	Fri 6/28/24	23
25	**User Documentation**	**14 days**	**Mon 5/26/25**	**Thu 6/12/25**	
26	Develop User Documentation	9 days	Mon 5/26/25	Thu 6/5/25	50
27	User Documentation Approval	5 days	Fri 6/6/25	Thu 6/12/25	26
28	**Designing and Prototyping**	**80 days**	**Mon 7/1/24**	**Fri 10/18/24**	
29	**Product Designs**	**80 days**	**Mon 7/1/24**	**Fri 10/18/24**	
30	Develop Initial Prototype	45 days	Mon 7/1/24	Fri 8/30/24	21,24
31	Test and Iterate on Prototype	11 days	Mon 9/2/24	Mon 9/16/24	30
32	Gather User Feedback	5 days	Mon 9/16/24	Fri 9/20/24	31
33	Refine Design Based on Feedback	15 days	Mon 9/23/24	Fri 10/11/24	32
34	Approve and finalize	5 days	Mon 10/14/24	Fri 10/18/24	33
35	**Mobile App Design**	**55 days**	**Mon 7/1/24**	**Fri 9/13/24**	
36	Design Low fidelity Design	10 days	Mon 7/1/24	Fri 7/12/24	24
37	Design High fidelity Design	35 days	Mon 7/15/24	Fri 8/30/24	36
38	Review and Approve	10 days	Mon 9/2/24	Fri 9/13/24	37
39	**Execution**	**130 days**	**Mon 9/16/24**	**Fri 3/14/25**	
40	**Software Development**	**130 days**	**Mon 9/16/24**	**Fri 3/14/25**	



	Task Name	Duration	Start	Finish	Predecessors
34	Approve and finalize	5 days	Mon 10/14/24	Fri 10/18/24	33
35	**Mobile App Design**	**55 days**	**Mon 7/1/24**	**Fri 9/13/24**	
36	Design Low fidelity Design	10 days	Mon 7/1/24	Fri 7/12/24	24
37	Design High fidelity Design	35 days	Mon 7/15/24	Fri 8/30/24	36
38	Review and Approve	10 days	Mon 9/2/24	Fri 9/13/24	37
39	**Execution**	**105 days**	**Mon 10/21/24**	**Fri 3/14/25**	
40	**Software Development**	**105 days**	**Mon 10/21/24**	**Fri 3/14/25**	
41	Configure Software	15 days	Mon 10/21/24	Fri 11/8/24	33
42	Creating AI and ML Algorithm	20 days	Mon 10/7/24	Fri 11/1/24	24
43	Coding	65 days	Mon 11/4/24	Fri 1/31/25	21,24
44	Integration with prototype and mobile app	20 days	Mon 2/3/25	Fri 2/28/25	38
45	Unit Testing	10 days	Mon 3/3/25	Fri 3/14/25	44
46	**Integration and Testing**	**77 days**	**Mon 3/17/25**	**Tue 7/1/25**	
47	**System Test Plan**	**77 days**	**Mon 3/17/25**	**Tue 7/1/25**	
48	Conduct Internal testing	20 days	Mon 3/17/25	Fri 4/11/25	45
49	Perform Integration Testing	15 days	Mon 4/14/25	Fri 5/2/25	48
50	Acceptance Testing	15 days	Mon 5/5/25	Fri 5/23/25	49
51	Address Bugs and Issues	26 days	Mon 5/26/25	Mon 6/30/25	50
52	Go live	1 day	Tue 7/1/25	Tue 7/1/25	51
53	**Implementation and Support**				
54	Support				

Figure: Gantt Chart of Ray-Ban Meta Smart Glasses Project

PART E : RISK REGISTRY

No	Rank	Risk	Description	Category	Root Cause	Triggers	Potential Responses	Risk Owner	Probability	Impact
1	1	Cybersecurity Threats	Fraudulent attempts to acquire sensitive information through trickery for cybersecurity threats.	Cybersecurity	Employees not being well-informed, and emails not being strongly protected.	More tricky emails trying to deceive, and passwords being at risk.	Provide frequent cybersecurity education, put sophisticated email filtering in place, and mandate multi-factor authentication	Chief Information Security Officer	High	High
2	2	Supply Chain Risks	Factors impacting suppliers, resulting in delays or shortages.	Supply Chain	vulnerabilities in the supply chain network leading to disruptions.	Suppliers closing, and difficulties with transportation.	Implement backup plans, expand your supplier base, and keep an eye on world events	Supply Chain Manager	High	High
3	3	Compliance and Regulation	Inability to provide required documents which include valid license and proper certificates from respective authorities	Legal Risk	Not complying with the new regulations or standards.	When a new bill or law is passed changing the existing regulations.	Develop a comprehensive legal document that follows all the regulations and follow their requirements and stay updated with the new released laws	The legal and regulation compliance team	High	High
4	4	Schedule and Budget Overruns – Finance risk	Risk of not being able to fund the anticipated amount for the project	Financial	Inability to raise the necessary funds for the project	Increase in raw material costs or operational costs	Accounts team to work with the investors and stake holders	Accounting Team	Low	Med

5	5	Software Development	Software code not working as intended and failure to provide accurate data	Software Technology	Inaccurate data and insufficient software testing without following proper procedures.	Insufficient human and software tools to create bug free code results, not following strong security practices.	Implement regular software testing procedures to provide frequent updates that follow standard policies.	Code development team	Medium	High
6	6	Marketing and Launch	Poor marketing campaigns can lead to low customer interest.	Marketing Risk	Insufficient grasp of the market and lack of alignment.	Insufficient research into the market and misunderstanding trends.	Analyze marketing tactics again, carry out more market research, and modify campaigns	Marketing Team	Medium	High
7	7	User Acceptance : User Experience (UX), User Interface (UI) design	Customers are unable to adapt to the complex software and poor interface design.	Technical Risk	Customer dissatisfaction when using the product.	Difficulty in using the product and lack of research on user requirements. Inconsistency in UI/UX design.	Establish easy and proper design guidelines to ensure Every customer can use the product.	UI /UX Designer, Team Lead	Low	High
8	8	Data Security and Privacy	Leading to the exposure of sensitive information can result in a potential breach.	Data Security	Insufficient security measures, lack of encryption	Weak authentication, hacking attempts	Boost security procedures, add multi-factor authentication, and encrypt important information	Chief Information Security Officer	Low	High
9	9	Technical Challenges – Prototype Failure	Failure of developing a prototype with all the business requirements	Technical risk	Lack of expertise	Lack of proper development would lead to a non-efficient prototype	Reach out to the respective patent owner for expertise advice	Research and Development team	High	High

10	10	Connectivity / Integration	Connectivity issue due to limited wi-fi range or faulty Bluetooth module.	Information Technology	Using faulty components or designing a proper antenna to transfer data over two parties.	Not equipped with the latest modules to support new versions and not testing in different environments to get feedback.	Optimize antenna design for high range and regular software updates to support Wi-Fi and Bluetooth modules.	Software team, Product manager, Design team.	Medium	Medium
11	11	Health and Fitness Monitoring	Compromising user privacy, and unauthorized access to health data is a concern	Security	Inadequate safeguards for data, such as lacking codes and protective measures.	Online attack by exploiting system weaknesses.	Improve data encryption, put multi-factor authentication into place, and carry out frequent security assessments	IT Security Team	Medium	Medium
12	12	Durability and Comfort	The smart glasses lifespan is affected by their durability and users experience discomfort when using the glasses.	Development Risk	Fragile structure and rigorous daily usage.	Accidental drops in water or harsh environment, discomfort or eyes straining when using the glasses for a longer time.	Analyze the user feedback and re-create the glass frame to match their satisfaction. Use better materials in manufacturing the smart glasses.	Product design Quality assurance team,	Low	Medium
13	13	Hardware Development	Hardware components failure due to manufacturing defect	Hardware Technology	Proper circuit design to thermal management and fluctuations in current flow and components assembly defects	Using Low quality componentss to reduce material cost and not reviewing quality design	Implementing Quality testing during the manufacturing phase and using proper safeguard methods of individual component failure.	Manufacturing and Quality assurance team	Low	High

14	14	Battery Management	Frequent battery discharging and low battery life.	Operational Risk	Frequenting recharging the smart glasses and faulty battery recharging module	Using the product for a longer duration or defective battery equipment	Create a new power management procedure to maximize battery efficiency and withstand longer duration and Ensure batteries are sealed properly.	Battery design team, Team lead	Low	Low
15	15	Competition – Product Duplicate	Risk of similar or duplicate product in market	Market	Similar product with a lower cost in market	Copyright infringement by low-cost manufacturers	Legal team to sue the copyrights	Legal Team	Low	High

Table: Risk Register for Ray-Ban- Meta Smart Glasses

PART E.1 : RISK IMPACT MATRIX

A risk matrix can help businesses cultivate a solid understanding of the risk environment, helping them manage and mitigate risks before they occur. The magnitude and complexity of business risks continue to grow.

PROBABILITY	LOW	MEDIUM	HIGH
HIGH		Customer support and maintenance [10].	Cyber security threat [1]. Supply chain Risk [2]. Compliance and regulations [3]. Technical Challenges – Prototype Failure [9].
MEDIUM		Connectivity [10], Health and Fitness monitoring [11]	Marketing and Launch [6]
LOW	Battery Risk [14]	Durability and comfort [12]. Schedule and Budget Overruns – Finance risk [4].	User Acceptance : User Experience (UX), User Interface (UI) design [7]. Data Security and Privacy [8]. Software Development [5], Hardware Development [13]. Competition – Product Duplicate [15].

IMPACT

Figure: Impact Matrix of Ray-Ban Meta Smart Glasses

PART F: REFERENCES

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